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Palaeomagnetic studies within the Ballantrae Ophiolite; southwest Scotland: magnetotectonic and regional tectonic implications

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The Early Arenig Slockenray Formation within the Ballantrae Ophiolite, southwest Scotland, displays a multivectorial magnetisation structure. Two components (S and M) are identified delineated by differing blocking temperature/coercivity spectra. Component S is removed around 200 °C/10 mT, and is regarded to be of recent viscous origin. Component M forms the characteristic formation magnetisation and resides in both magnetite and haematite. Extensive sampling of all exposed lithologies reveals an (in situ) non-Fisherian distribution of the characteristic magnetisation defining an envelope from SE moderate positive to SW shallow negative directions. A negative infra-formation conglomerate test identifies this component as a pervasive overprint.

A second conglomerate test performed in the overlying Benan Conglomerate of Llandeilo age, reveals dispersely directed magnetisation with a stability range equivalent to that of component M. This field test therefore defines a maximum remagnetisation window of 30 million years for the characteristic remanence. "Hard" viscous magnetisations are identified both in the Benan Conglomerate and at some sites within the Slockenray Formation.

Structurally corrected site mean results from the Slockenray Formation define a non-Fisherian distribution and form a small circle partial arc centred on a vertical axis (NW moderate positive to SW moderate positive directions). A combined palaeomagnetic fold and fault test suggests that acquisition of component M pre-dates both folding and faulting.

The resulting palaeolatitude of remanence acquisition (28.8 °S) implies a tectonic position close to the southern Laurentian margin for the Ballantrae ophiolite in Arenig times.

1. Introduction and geological background

The Early Ordovician Ballantrae Ophiolite is exposed over an area of approximately eighty square kilometres on the Ayrshire coast of southwest Scotland. The ophiolite comprises a heterogenous assemblage of oceanic and mantle derived lithologies which have become structurally juxtaposed during obduction [1]. The outcrop area is dominated by two SW-NE trending tracts of ultramafic rocks and spilites referred to as the southern and northern belts of the ophiolite (Fig. 1). The ophiolitic character of the complex was first recognised by Church and Gayer [2] and Dewey [3]. Subsequently, the ophiolite has played an important role in the moulding of tectonic models for Caledonian Britain. The original tectonic setting is controversial, however, and various authors invoke ocean island, marginal basin and oceanic ridge settings for ocean crust generation (cf. [4–6]).

The present palaeomagnetic study focuses on the Slockenray Formation of the northern belt. This unit is comprised of a sequence of spilitic lavas together with a variety of intercalated sediments. The present investigation considers samples from all the exposed lithologies of the Slockenray Formation including an intra-formational conglomeratic facies. The sampled area is bounded on the south side by a major fracture and to the north by an unconformable younger sedimentary sequence (Fig. 1). Large-scale outcrop maps [5] were used to constrain stratigraphic, structural and geographic control for the palaeomagnetic sampling sites.

An additional palaeomagnetic conglomerate test was performed in the Benan Conglomerate of mixed source provenance containing volcanic clasts likely derived from the Ballantrae Ophiolite. This conglomerate forms part of a proximal forearc assemblage founded on the obducted ophiolite complex [7]. Fig. 1 shows the geographic and

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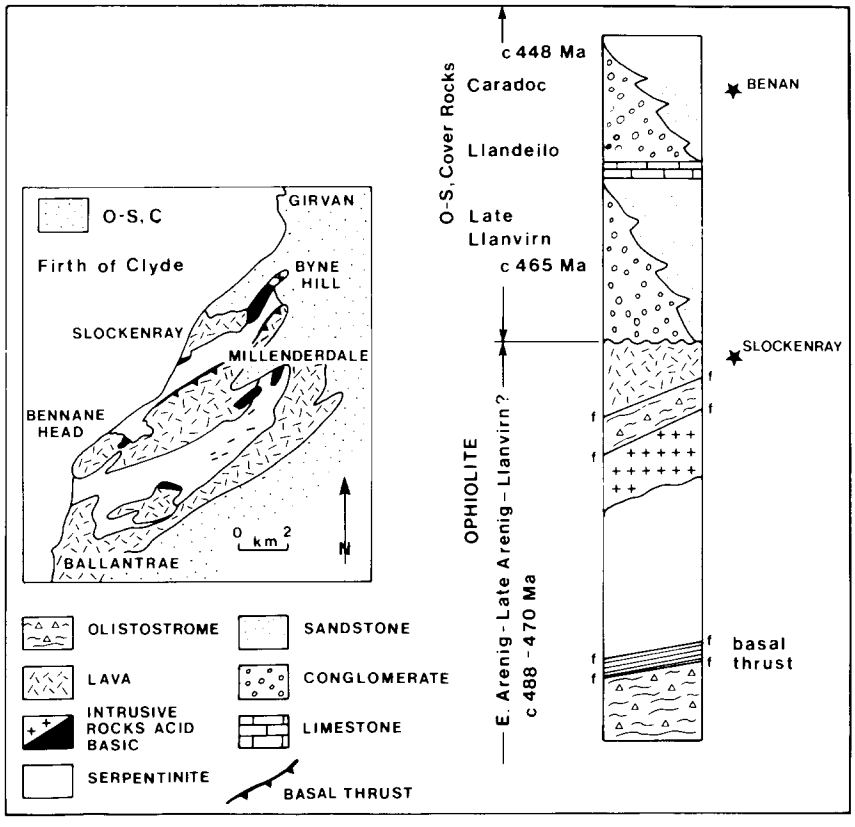


Fig. 1. Geological sketch map and stratigraphic vertical section of the Ballantrae area. Sampled conglomerates are starred. *f* = faulted contact., *O-S* = Ordovician/Silurian cover rocks.

stratigraphic positions of the Slockenray Formation and Benan Conglomerate.

The Slockenray Formation is Arenig in age [8,9] which accords well with radiometric determinations elsewhere in the ophiolite summarised by Hamilton et al. [10]. Macrofossil evidence from a lateral equivalent of the overlying Benan Conglomerate dates the unit as of Llandeilo age [11,12].

Previous palaeomagnetic results from the ophiolite (i.e. [13,14]) have been incorporated with other British Ordovician data in apparent polar wander schemes (e.g. [15,16]). This carries the assumption of a post-obduction origin for the recovered magnetisations as no accretion related deformation was accounted for.

The present study was therefore aimed at the uncovering of "primary" or pre-obduction magnetisations using more detailed demagnetisation analyses compared with these earlier studies. Also the availability of a detailed map of this part of the complex affords the opportunity to deal with structural complexity using the palaeomagnetic data interactively. It was hoped that constraints might then be placed with respect to the tectonic evolution of the ophiolite in the context of the British Caledonides.

2. Sampling and methods

Palaeomagnetic sampling sites were chosen to incorporate all the exposed lithologies of the Slockenray Formation. These included sheet lavas, pillow lavas, shale, chert, lithic arenites and lithic tuffs (15 sites, 97 samples; Table 1). Both intra- and extra-formational conglomerates were sampled to test the stability of identified magnetisation components (Slockenray Conglomerate, 19 clasts; Benan Conglomerate, 21 clasts).

A portable petrol-driven drill was used to collect 2.5 cm diameter samples oriented using both magnetic and solar compass wherever possible. All sites were located on large-scale outcrop maps (Fig. 2) and comprised at least five independently oriented samples.

Natural remanent magnetisation (NRM) was measured on a fluxgate magnetometer [17]. Thermal and alternating field (AF) cleaning techniques were performed on samples in the approximate ratio 2 : 1. Thermal demagnetisation was carried out using two furnaces capable of holding a combined total of 18 samples. The present earth field was attenuated using mu-Metal shields surrounding each furnace. AF demagnetisation proceeded using a microcomputer-controlled static three-axis system working at 186 Hertz.

Demagnetisation results were plotted on

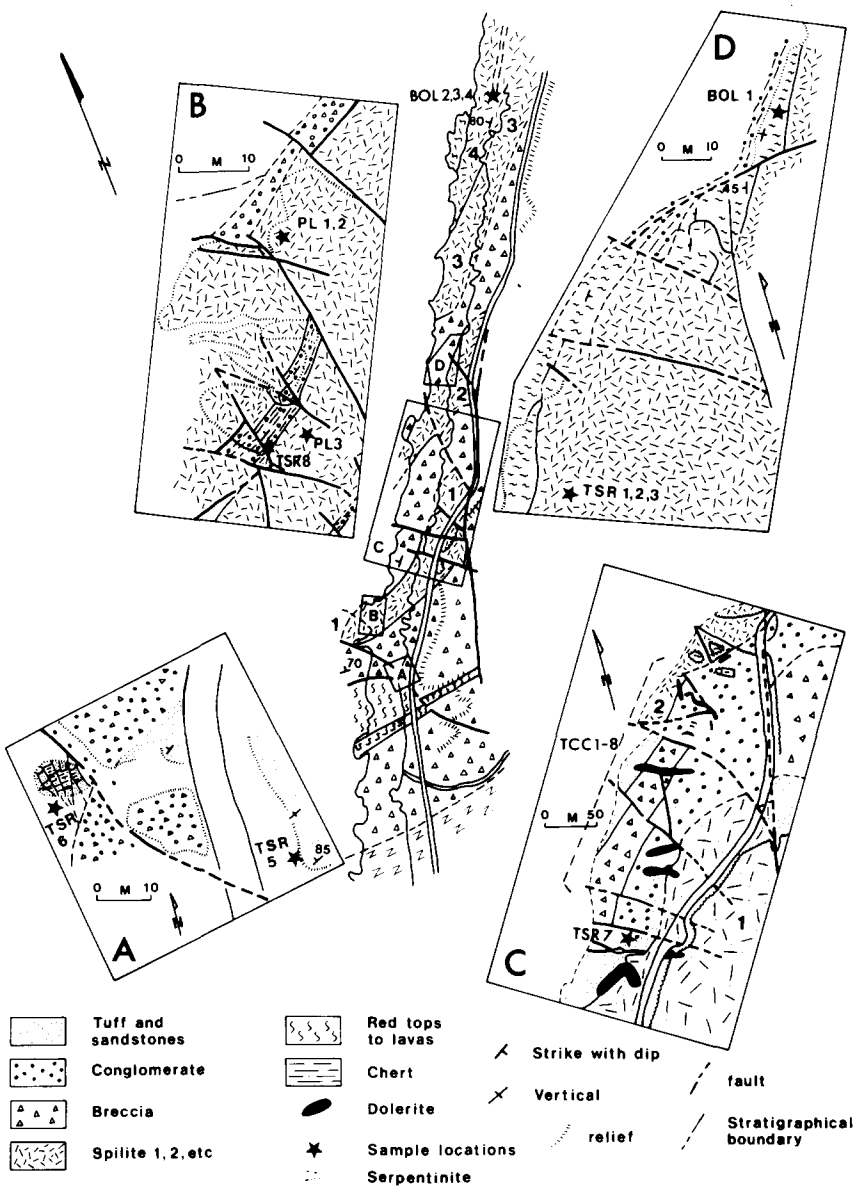


Fig. 2. Simplified large-scale outcrop maps of the Slockenray section show the distribution of palaeomagnetic sampling sites. Spilite numbers are those of Bluck [5].

TABLE 1

Site lithologies and site mean Fisher statistics for the Slockenray Formation (component M) and Benan Conglomerate

Lithology	Site	N	α_{95} (°)	K	In situ		Strike corrected		Plunge fold corrected	
					D (°)	I (°)	D (°)	I (°)	D (°)	I (°)
Porphyritic basalt	TSR1	5	9.7	62.8	150.1	43.6	301.5	48.6	278	48.6
Basalt	TSR2	6	9.7	49.0	155.7	26.9	279.8	60.7	258	60.7
Porphyritic basalt	TSR3	6	7.1	91.1	144.5	40.2	306.3	53.1	285	53.1
Black shale	TSR5	5	14.6	28.3	204.1	1.5	249.6	51.6	202	51.6
Chert	TSR6	6	19.9	12.3	210.1	11.4	276.3	49.3	222	49.3
Lithic arenite	TSR7	6	8.9	57.0	164.9	15.4	212.1	44.6	207	44.6
Lithic arenite	TSR8	6	11.8	33.1	176.3	−2.3	203.2	36.0	198	36.0
Pillow lava	PL1	11	6.9	44.2	167.2	10.2	209.3	56.1	207	56.1
Pillow lava	PL2	8	6.4	77.0	174.5	11.1	215.0	49.8	211	49.8
Pillow lava	PL3	7	12.0	26.4	188.2	4.5	213.8	29.4	202	29.4
Reddened lava	BOL1	6	12.7	28.6	206.6	9.6	222.7	9.3	216	9.3
Red lithic tuff	BOL2	7	18.4	11.7	128.4	27.8	265.9	53.4	258	53.4
Red lithic tuff	BOL3	7	13.4	21.3	126.7	51.6	279.2	39.8	278	39.8
Red lithic tuff	BOL4	5 [a]	4.9	241.5	150.3	76.8	283.0	15.2	282	15.2
Slockenray Conglomerate	TCC1–8	30 *,[a]	9.0	9.5	171.0	21.3				
		19 **	10.3	11.5	167.9	21.4				
Benan Conglomerate	BCT1–24	37 *	22.4	2.1	351.5	56.7				
		21 **	29.2	2.2	354.9	61.9				

Statistics are outlined both by sample (*) and by clast (**) for the conglomerate analyses.
Notation is as follows: N = number of samples; K = Fisher precision parameter; D = palaeomagnetic declination; I = palaeomagnetic inclination; α_{95} = cone of 95% confidence.
[a] Note: component M was not recovered from ten samples of clast TCC1 (Slockenray Conglomerate) or from two samples of site BOL4 (AF demagnetisation).

Zijderveld [18] diagrams/orthogonal projections and interpreted interactively utilising the LINEFIND statistical analysis algorithm [19].

3. Results of palaeomagnetic analysis

3.1. General

NRM intensities vary considerably between sample sites expectantly reflecting the variation in sampled lithology. In general, the volcanic samples have greater intensities than the intervening sediments, although sites of particularly weak intensity (< 5 mA/m) for both lithological realms were discovered (total range: 1 mA/m to 5 A/m). Laboratory cleaning techniques however, produced good results independent of initial NRM intensity throughout the study.

NRM directions are reasonably well grouped with southerly declinations and moderate positive inclinations. Generally, on cleaning, resulting directions tended towards shallower inclinations reflecting the removal of a low treatment present-day field component.

Both thermal and AF cleaning methods proved successful in resolving magnetisation components for the majority of site lithologies. Usually two components of magnetisation are resolved within a sample. A "soft" component (S), generally removed below 200 °C/10 mT, evolves gradually to a more stable component (M) decaying linearly to the origin by approximately 560 °C/60–100 mT. Typical examples of characteristic demagnetisation behaviour are illustrated in Fig. 3.

The soft component S aligns close to the present-day field (in situ) and is undoubtedly of VRM (viscous) origin. Component M varies in attitude (in situ) comprising a non-Fisherian distribution of directions with SE-SW declination and moderate positive to shallow negative inclinations (Table 1). This component forms the characteristic magnetisation of the formation and occurs in all samples; component S is sometimes absent or only

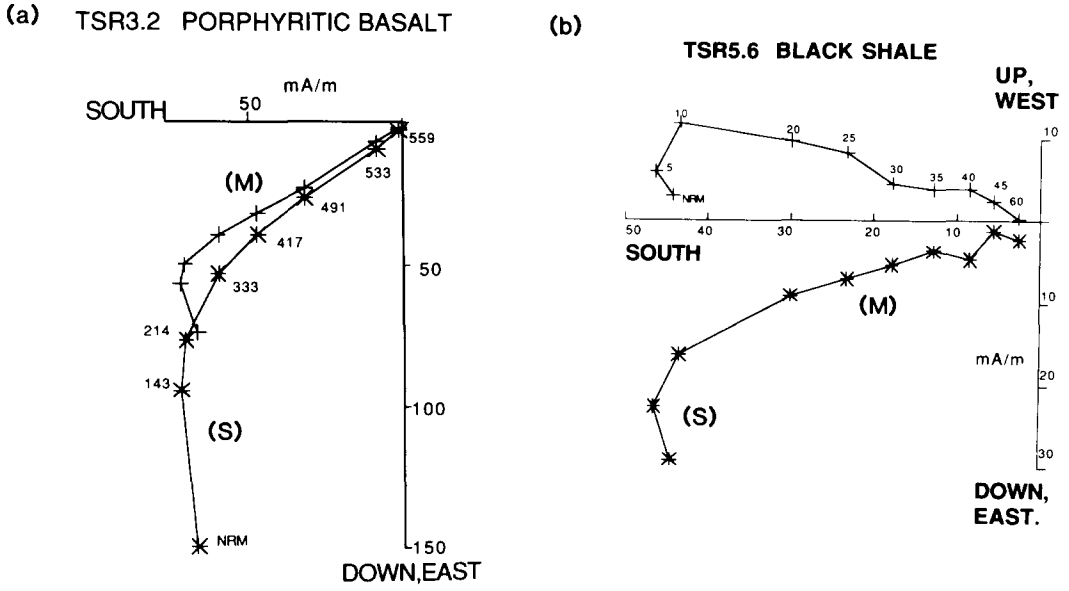


Fig. 3. Examples of orthogonal projections showing components S and M defined by both thermal (a) and AF (b) cleaning methods for the Slockenray Formation. Throughout this paper crossed symbols in the orthogonal vector diagram refer to points in the horizontal plane and starred symbols to points in the vertical plane. Thermal (AF) treatments are always given in degrees Celcius (milliTesla).

expressed as a minor "kink" of the orthogonal projection following the first demagnetisation step.

Exceptions to this typical demagnetisation behaviour are as follows:

Sites BOL1–BOL3. These sites comprise a reddened lava top (BOL1) and reddened lithic tuff units (BOL2, 3) within the Formation. Although true boles (palaeosoils) are not developed, Smellie [20] suggests subaerial weathering as a possible explanation for the reddening. Thermal and AF demagnetisation of these sites shows a dual component structure as for previous sites but the stability spectrum of component M is extended. Thermal demagnetisation produces two discrete unblocking temperatures of remanence in the 500–560 °C and > 600 °C thermal windows. Fig. 4a–c illustrates this behaviour for a sample from site BOL1 (BOL1.3). LINEFIND analysis confirms the decay of the magnetic vector to be univectorial spanning both unblocking spectra. AF demagnetisation of samples is incomplete at 100 mT indicating the presence of a high coercivity remanence phase (Fig. 4d).

Site BOL4. This site was also sampled in a reddened lithic tuff unit. Demagnetisation using both cleaning methods again reveals the presence of two components of magnetisation. The stability properties of components S and M, however, are markedly different from those previously described. Component S is found to predominate over component M in relative intensity at this site. Thermal demagnetisation removes only component S until temperatures exceeding 250 °C and curvature when viewed in orthogonal projection indicates considerable overlap of component stability to temperatures approaching 600 °C (Fig. 5a, b). AF demagnetisation is incomplete after treatment at 100 mT and reveals only component S. However, the existence of component M is demonstrated as the demagnetisation does not proceed to the origin (Fig. 5c, d).

Site TSR6. For several samples of the study, the LINEFIND statistical algorithm is unable to successfully include the origin within the stable magnetisation decay defined by component M. This effect is particularly evident for the cherts sam-

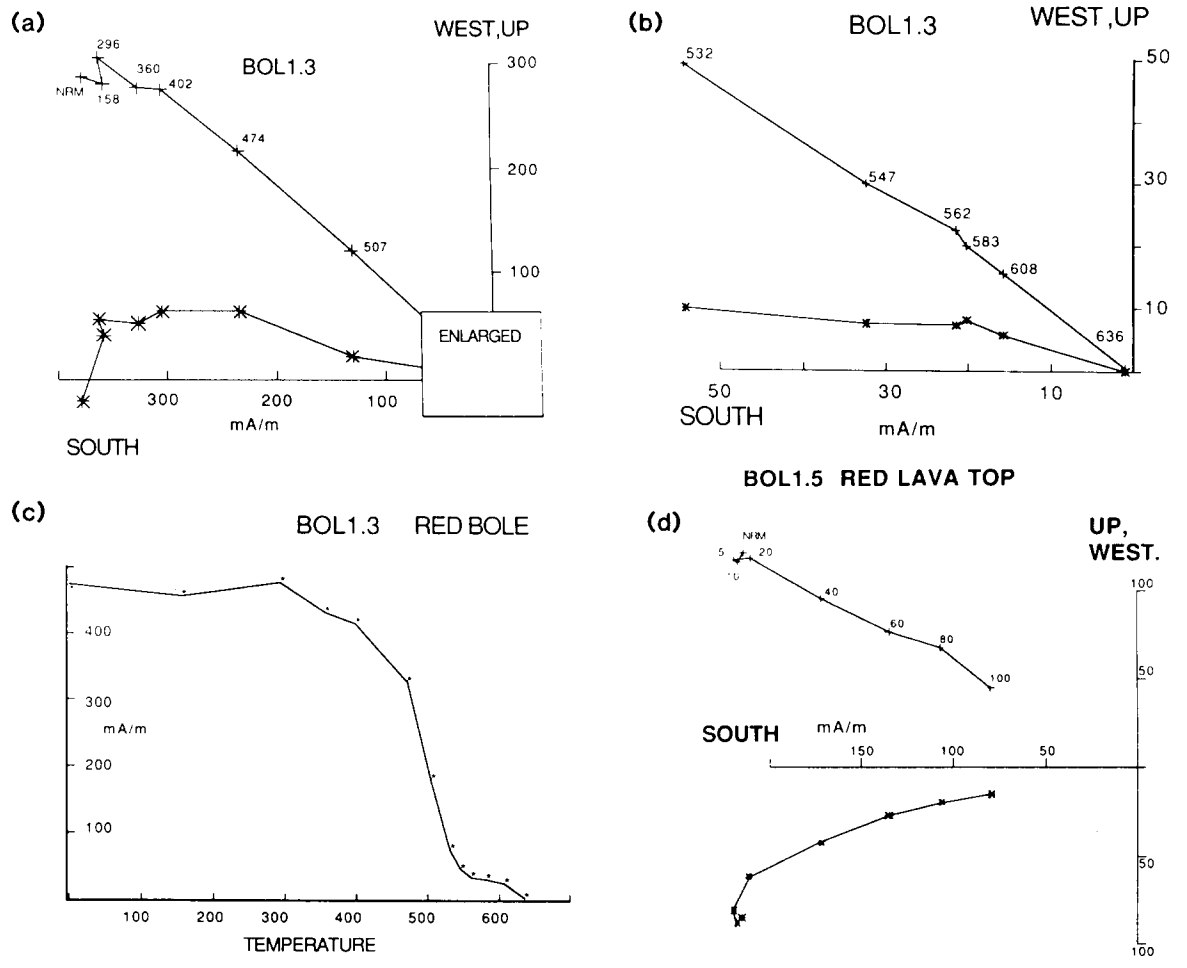


Fig. 4. Orthogonal projections (a, b, d) and intensity/temperature distribution (c) record the response of site BOL1 to thermal (a, b, c) and AF (d) cleaning. The high treatment results of sample BOL1.3 (a) have been enlarged (b) to illustrate their linear demagnetisation close to the origin.

pled (TSR6) for which component M visibly decays past the origin.

Possible explanations for this behaviour are discussed by Klootwijk [21] and are as follows:

— Inhomogenous magnetisation [22] which becomes more prominent as demagnetisation continues.

— Presence of an unidentified additional magnetic component.

— VRM acquisition due to a residual field within the magnetic shield of the thermal demagnetiser.

Although attempts were made to isolate any residual component present at high treatments, these proved unsuccessful due to the noise levels encountered near the origin during sample demagnetisation.

3.2. Conglomerate tests

Two conglomerate tests were performed within the course of the study as follows:

Slockenray Formation intra-formational conglomerate facies. The conglomeratic unit of the Slockenray Formation forms the proximal facies of a hyalotuff delta sourced from the Slockenray basalts themselves [5]. A total of forty samples were collected from nineteen clasts for laboratory analysis. NRM directions closely agree with those obtained from the parent flows (southerly declina-

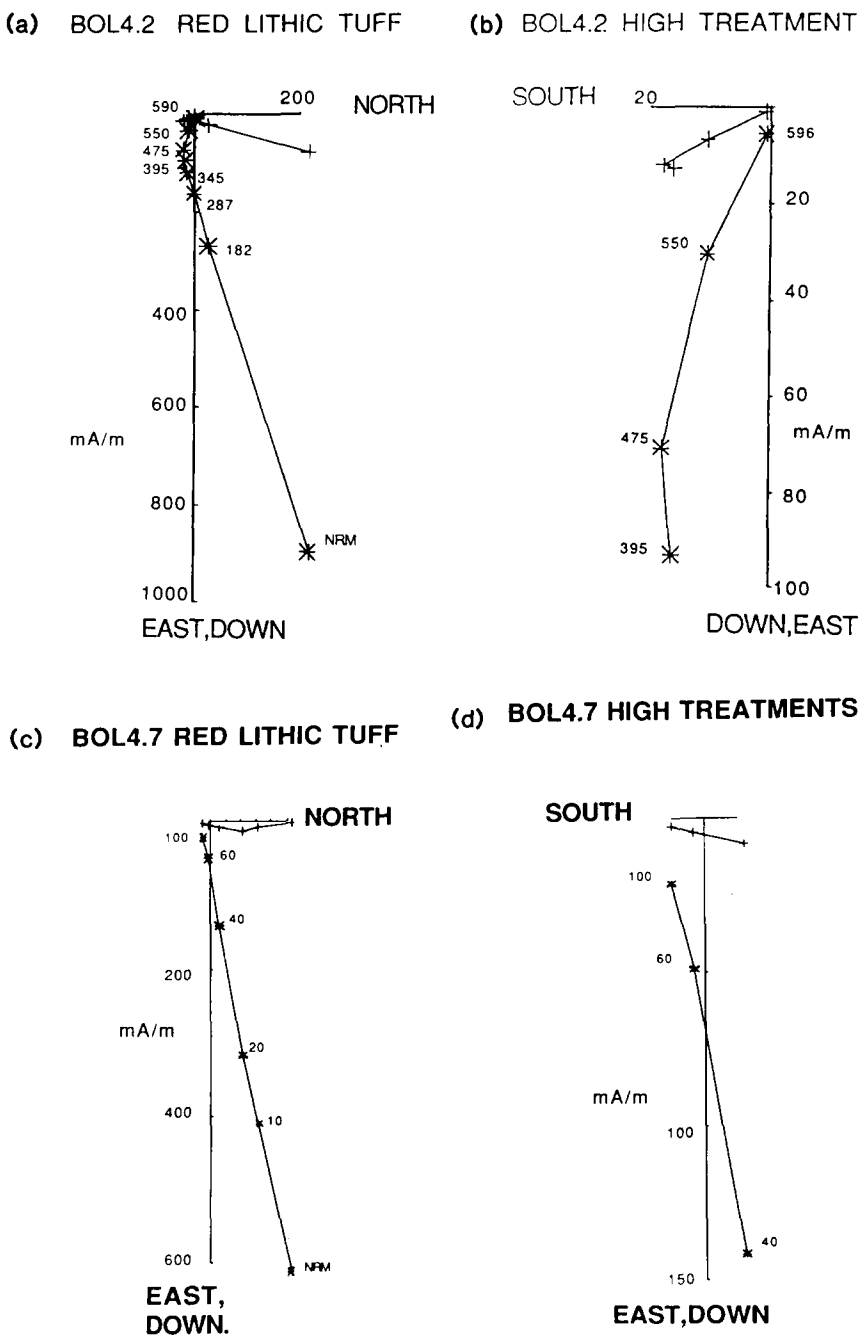


Fig. 5. Orthogonal projections record the demagnetisation behaviour of site BOL4 to thermal (a, b) and AF (c, d) cleaning. Enlarged projections (b, d) are included to show the dual component nature of remanence revealed close to the origin.

tion, moderate positive inclination). Intensities of magnetisation from the conglomerate clasts are slightly higher than those observed in the parent flows but are of the same order.

A two-component magnetisation signature characterises clasts and is analogous to that observed for the rest of the Slockenray Formation. A "soft" component (S) with steep inclination and

generally northward declination is removed below 250 °C/10 mT. This leaves a more stable component (M) of southerly declination and moderate positive inclination which decays linearly to the origin following treatment to 560 °C/100 mT. In situ stereographic projection of component M and representative orthogonal projections for the Slockenray Conglomerate are shown in Fig. 6. Statistical analysis of component M from the conglomerate is included in Table 1.

Atypical behaviour to that described above occurred in only one sampled clast (TCC1). Eleven samples were cored from this clast of which only one showed the characteristic demagnetisation be-

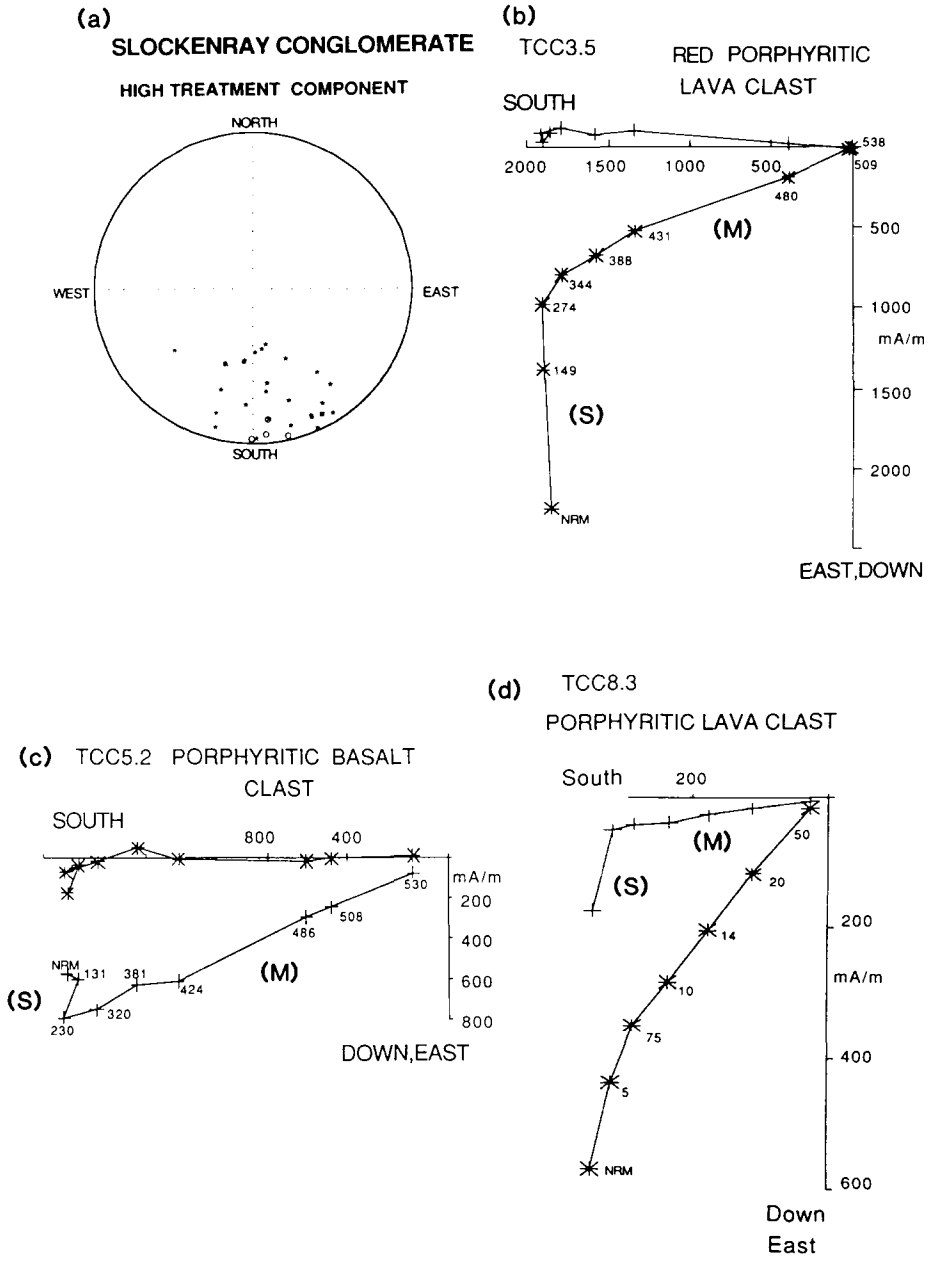


Fig. 6. (a) Directional distribution of component M magnetisations for the intra-formational Slockenray Conglomerate. Thermal (b, c) and AF (d) orthogonal projections show typical demagnetisation trajectories.

haviour. The remainder of the samples appeared to contain only component S which was of unusually high stability decaying linearly to the origin by 550°C/65 mT.

Benan Conglomerate. The Benan Conglomerate has a dual provenance origin and comprises lithologies of the underlying ophiolite and those of an eroded arc terrane [7]. Thirty-seven samples were collected from twenty-one clasts of the conglomerate exposed directly above the ophiolite at Byne Hill (Fig. 1). NRM directions showed a widely dispersed distribution but with some clustering close to the present field orientation at the site. NRM intensities covered a broad range similar to those determined within the ophiolite (1 mA/m to 5 A/m). Many clasts were found to display the "soft" magnetic component (S) again aligned (in

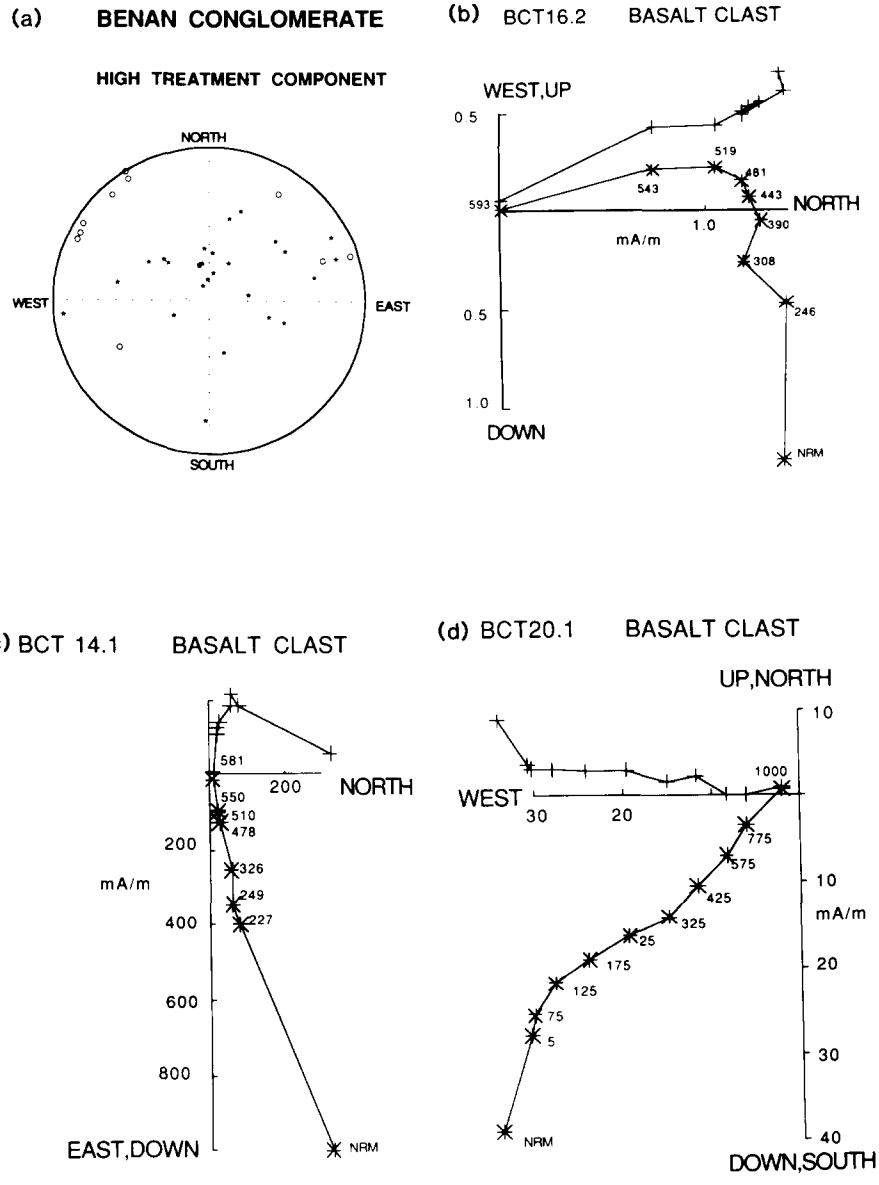


Fig. 7. (a) Directional distribution of cleaned magnetisation components from the Benan Conglomerate. Typical thermal (b, c) and AF (d) orthogonal projections show the directionally dispersed nature of the high treatment component magnetisations.

situ), close to a present-day field orientation. Magnetisation components of similar stability properties to component M do not display a singular orientation however. In situ stereographic projection of these "high treatment" components and typical orthogonal projections for basalt clasts from the Benan Conglomerate are shown in Fig. 7.

3.3. Palaeomagnetic fold and fault tests

The Slockenray Formation has a general NE-SW strike younging (where determinable) toward the northwest. The observed dips vary from steep to subvertical (Fig. 2).

Both original depositional and tectonic components of dip are identified [5] consistent with the interpretation that the unit formed at a palaeo-shoreline. Large-scaled foresets, identified at sites TCC1–8 (Slockenray Conglomerate), have not been structurally corrected. Elsewhere, observed bedding and/or lava tops have been taken as representative of the palaeo-horizontal. Despite the broad similarity in bedding attitude throughout the sequence, a comparison of pre- and post-structural correction site mean distributions has been made in an attempt to relate magnetisation age to deformation. Furthermore, comparison of magnetisation directions between small-scale fault blocks (identified from outcrop maps; Fig. 2) has been performed. This allows the relation of magnetisation age to fault deformation.

Fold test. Poles to bedding from each sampling locality form a partial great circle distribution (Fig. 8). The pole to this plane has declination 290°, inclination 80° and represents the axis of a plunging fold structure affecting the Slockenray Formation. The structural correction of the palaeomagnetic data may therefore be problematical. Two differing tectonic corrections have therefore been applied to the in situ distribution of the characteristic magnetisation component (M).

Firstly, tilt can be assumed to have taken place about a horizontal strike axis at each site and restored using the conventional tilt correction. Secondly, sequential unplunging and unfolding of the determined fold axis can be performed. The effects of these two contrasting tectonic corrections applied to the Slockenray data are illustrated in Fig. 9.

The in situ site means form a circular swathe

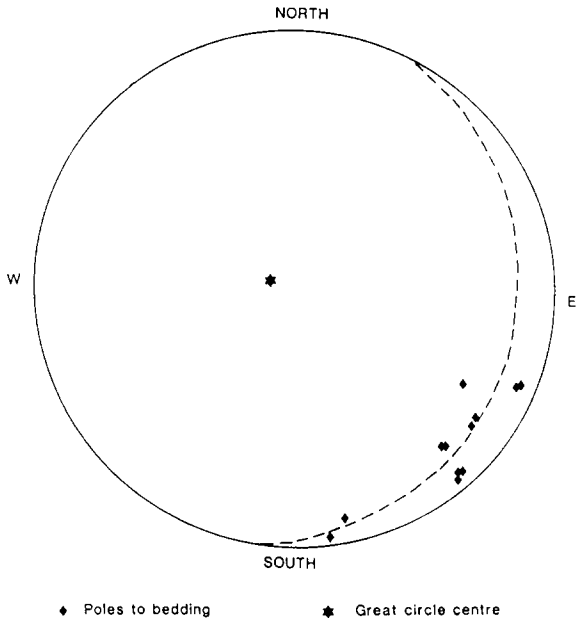


Fig. 8. Equal angle stereographic projection shows the partial great circle distribution of poles to bedding for palaeomagnetic sampling sites.

which varies in both declination and inclination (Fig. 9a). By contrast, the structurally corrected site mean distributions show considerably less variation in inclination but remain dispersed in declination (Fig. 9b, c). Exceptions to this relationship are the mean results from sites BOL1 and BOL4 which appear markedly shallow in inclination.

Fault test. Numerous small-scale faults transect the study area as evident from Fig. 2. A comparison of magnetisation directions both within and between fault blocks has been carried out and is illustrated in Fig. 9 (consistent ornament denotes sampling sites from within individual blocks). Fault blocks themselves were defined as bound by discontinuities of sizable or unknown displacement. Within site minor offsets of decimetre to metre scale were therefore ignored.

Component M site means are found to overlap at the 95% confidence level within blocks, but significant departures are apparent when results are compared between fault blocks. This relationship appears to be valid both for in situ and corrected site mean distributions. Once again site BOL4 is an exception proving statistically differ-

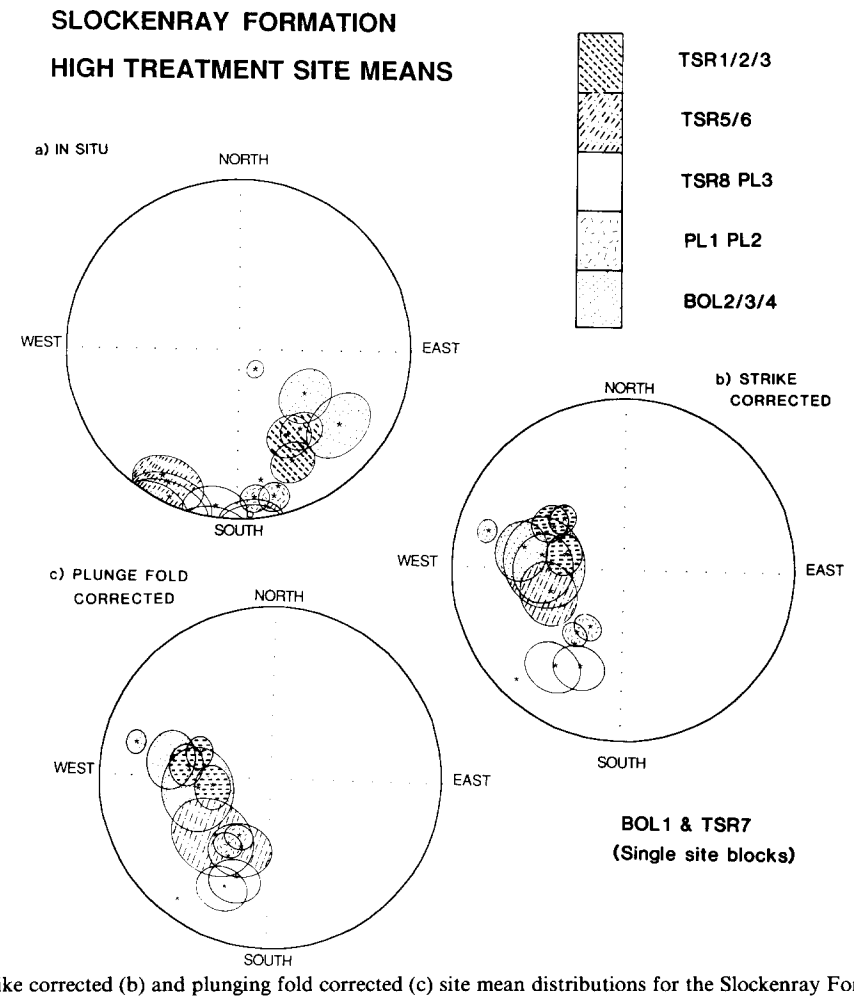


Fig. 9. In situ (a), strike corrected (b) and plunging fold corrected (c) site mean distributions for the Slockenray Formation plotted in equal area stereographic projection. Ovals depict cones of 95% confidence for each site mean. Intra-fault block sites are denoted by consistent ornamentation. Confidence cones for single site blocks have been omitted.

ent from other intra-block sites (BOL2, 3) at the 95% confidence level.

4. Interpretation of palaeomagnetic results

4.1. Identification of remanence phases

Maximum unblocking temperatures near 560 °C for the characteristic magnetisation component M at most sites is consistent with a magnetite–low-Ti titanomagnetite held remanence. Persistence to temperatures in excess of 600 °C for the reddened lava top (BOL1) and reddened lithic tuffs (BOL2–4) indicates that haematite is an important remanence carrier at these sites, however. Incomplete demagnetisation following cleaning to 100 mT confirms the presence of high-coercivity haematite. Both results are in accordance with reddening resulting from subaerial exposure of the lava/tuff sequence at or soon after extrusion/deposition [20].

The preservation of primitive magnetic property variations when compared to modern ocean basalts both with respect to pillow lava geometry and to lithological contact relations (Trench, in preparation) favours a lack of post-magnetisation alteration affects to the observed magnetic phases. The affirmed univectorial nature of both magnetite and haematite demagnetisation (Fig. 4) suggests their remanence to have been acquired synchronously.

4.2. Conglomerate tests and remagnetisation age constraint

Magnetisation components of comparable stability spectra have been identified in both the Slockenray Conglomerate facies and the overlying Benan Conglomerate. In the former case, the characteristic component (M) is of similar attitude to that found in the remainder of the Formation (Table 1). This forms a negative palaeomagnetic conglomerate test with respect to component M and identifies it as a pervasive remanence overprint.

The "soft" magnetisation component (S) aligns along the present-day field direction for both conglomerates thus indicating a present/recent VRM origin.

For the Benan Conglomerate, a magnetisation component of similar stability criterion to component M displays no evidence of pervasive overprinting compatible with that for the Slockenray Conglomerate (Fig. 7). This constitutes a positive conglomerate test and therefore defines a remagnetisation age window for the underlying Slockenray Formation. Maximum width of this window is approximately 30 Ma when the assigned ages of the conglomerates are considered (Base Arenig–Top Llandeilo, 488–458Ma [23]).

Directions from the Benan Conglomerate are not completely random in orientation however owing to the presence of "hard" VRM in some of the clasts. This phenomenon would also account for the demagnetisation behaviour of clast TCC1 of the Slockenray Conglomerate (section 3.2) and explain the change in stability spectra of components S and M for site BOL 4 (Fig. 5).

Hard viscous remanence has previously been noted in ophiolitic rocks [24], where it occurs in intimate association with rocks containing VRM of "normal" stability. The "hard" viscous magnetization reported here appears resistant to both thermal and AF cleaning methods (Fig. 5).

The overprinted magnetisation component M ubiquitously resides in magnetite for clasts of the Slockenray Conglomerate. Attempts to follow demagnetisation to haematite unblocking temperatures proved unsuccessful despite the sampling of reddened basalt clasts (see Fig. 6b). The problem of whether both magnetic phases are overprinted therefore remains.

Fig. 4 clearly shows both magnetite and haematite remanences to be univectorial and implies synchronous magnetisation (section 4.1). Metamorphism of the lava pile is to low grade (zeolite to prehnite-pumpellyite facies [25]) and may reflect operation of a submarine hydrothermal system [26]. Maximum temperatures are therefore insufficient to cause pure thermal remagnetisation using the unblocking criterion of Pullaiah et al. [27], but may result in at least resetting of magnetite applying the alternative theory of Walton [28].

Thermal (or thermo-chemical) magnetite remagnetisation by hydrothermal circulation sufficiently apace after lava formation as to preserve directional consistency between magnetic phases is therefore suggested. Such an interpretation is consistent with the differential application of unblocking theories proposed by Kent and Miller [29]. They invoke the Walton theory to be applicable to multidomain magnetite but the Pullaiah theory to relate in the case of single-domain haematite. The use of either theory in the event of thermo-chemical remagnetisation is questionable however.

4.3. Relationship of component M to bedding attitude and small-scale faulting

We have ascertained (section 4.2) that the characteristic magnetisation component (M) identified in the Slockenray Formation represents a magnetic overprint; i.e. it post-dates the deposition of the intra-formational Slockenray Conglomerate. There remains the problem of deducing the temporal relationship of this magnetisation to the observed fold and small-scale fault deformations however. The ages of these deformations are poorly constrained, and may relate to either ophiolite obduction, subsequent phases of Caledonian compression, or in the case of the faulting, even to opening of the present Atlantic.

The polyphase structural history described by Williams [30] for the Ballantrae cover sequence is not readily recognisable within the rocks of the ophiolite basement. Furthermore, Piper [14] argues that Caledonian movements did not effect major refolding in the ophiolite based on the directional consistency of remanence data from the northern serpentinite belt. The weight of evidence therefore suggests that the bulk of folding relates to the assembling and obduction of the ophiolite [9],

with the age of small-scale faulting remaining uncertain.

Neither the "in situ" nor the "corrected" site mean distributions from Slockenray are Fisherian (Fig. 9) and therefore a meaningful application of the classical fold test [31,32] is not possible. Inspection of Fig. 9 reveals that whilst "in situ" results vary both in declination and inclination; "corrected" results vary predominantly in declination only. This behaviour can result from the incorrect assumption that folding of the magnetisation took place about the local strike direction [33] producing the effect of apparent tectonic rotation about local vertical axes. A corrected pre-folding magnetisation will then form a small circle partial arc centred on a vertical axis (cf. Fig. 9b, c).

In the present case, localised fault blocks between which true tectonic rotation may have taken place have been identified during plane table mapping. Statistically significant directional differences become apparent between blocks both before and after tectonic correction backed up by internal consistency of directions within individual blocks. It is therefore not possible to fully evaluate the relative contributions of true and apparent tectonic rotation to the remaining small circle partial arc distribution following tectonic correction. The existence of this vertically centred small circle does strongly suggest component M to be of pre-folding origin however. Similarly the existence of statistically significant directional differences not within but between fault blocks would favour a pre-faulting age for the component.

Two site mean results (BOL1, BOL4) appear to disobey the described relationship in that neither lies on the small-circle partial arc described by the remainder of the sampling site means following tectonic correction. In addition, site BOL4 remains statistically different from its corresponding intra-block sites (BOL2, BOL3).

The explanation of this latter disparity becomes evident when the previously described stability properties of components S and M at site BOL4 are considered (Fig. 5). Note also that in Fig. 9a, the BOL4 site mean lies close to a remagnetisation circle between a present Earth's field direction (PEF) and the in situ palaeo-field direction determined from the other sites of the particular block. Therefore, although the LINEFIND algorithm detects linearity in the region 550 °C–origin, this is in fact the result of blocking temperature overlap between components S and M. The true orientation of component M for site BOL4 is therefore never attained but is expected to be close to those of sites BOL2 and 3.

For site BOL1, there are unfortunately no intra-fault block sites with which to draw direct comparison. The demagnetisation characteristics for the site reveal the presence of both magnetite and haematite remanence phases (Fig. 4) and do not appear in any way "anomalous" other than in direction following tectonic correction. This suggests that the applied structural correction for the site may be in error and implies that the measured lava top may not accurately record the palaeohorizontal. Such problems may well have been predicted given the prevalence of syn-depositional dip components in the hyalotuff delta environment [5]. For these reasons, both sites BOL1 and 4 have been omitted from the overall statistical analyses.

5. Comparison of results to previous palaeomagnetic work

5.1. Slockenray lavas

Two previous palaeomagnetic investigations have been performed which include data from the Slockenray lavas; those of Nesbitt [13] and Piper [14]. The present work improves on both previous studies on a number of counts as follows:

— the use of large-scale plane table outcrop maps enabling the relation of remanence to small-scale fault blocks; a form of palaeomagnetic "fault" test;

— the undertaking of two conglomerate tests;

— the sampling of the interbedded sediments in addition to the lavas;

— the use of detailed thermal demagnetisation in conjunction with the previously used AF method.

Nesbitt [13] reports results from twelve hand samples collected from four sites in the Slockenray lava sequence. Following AF cleaning at 8.5 mT, a better clustering of sample vectors prior to tilt correction was described. This was interpreted as possibly reflecting an Ordovician overprint direction at declination 189 ° and inclination +40 ° for the formation. In the light of the present evidence,

this explanation becomes suspect given the apparent effects of later faulting and the arguments for a pre-deformational magnetisation. It may also be debatable whether cleaning at fields of 8.5 mT will be sufficient to completely remove presumed "soft" magnetisation components in some cases (cf. Fig. 5).

Piper [14] states that the Slockenray pillow lavas "did not yield an overall significant direction" and attributes this to "variable amounts of remagnetisation during folding and metamorphism". In contrast, the current study presents evidence for a pervasive remagnetisation (at least of the magnetite held remanence) as evidenced from the negative Slockenray conglomerate test (Fig. 6). The problems in determining "an overall significant direction" therefore result from the dispersion caused by both fold and fault deformations.

5.2. Byne Hill Gabbro and northern belt serpentinite [14]

Piper [14] reports palaeomagnetic data from eight sites (41 specimens) in the Byne Hill Gabbro and nine sites (57 specimens) from serpentinites in the northern belt. Directions result largely from AF demagnetisation complemented by those from a limited number of thermally demagnetised pilot specimens.

Site mean results from this study are plotted in Fig. 10. The directional distribution from the Byne Hill Gabbro is clearly non-Fisherian, and smeared along a NW-SE oriented great circle. This may result from either incomplete resolution of magnetisation components (compare sites BOL4 vs. BOL2/3, Fig. 9a) or less likely, may reflect unresolved structure affecting the gabbro. In either eventuality, the calculation of a Fisher mean and its interpretation as representative of a single Ordovician palaeofield becomes suspect. The northern belt serpentinite data form a better approximation to the Fisherian distribution and yield a mean direction of declination 147.7°, inclination +32.4°. This magnetisation was similarly attributed an Ordovician age and probably relates to magnetite generation during serpentinisation. Serpentinisation itself took place during late syn- to post-obduction of the ophiolite as evidenced through its geochemical influence on Arenig dyke suites cross-cutting the metamorphic sole [34].

The possibility that the Byne Hill Gabbro result does not represent an Ordovician field would remove one of the "large error" determinations currently problematic for the statistically signifi-

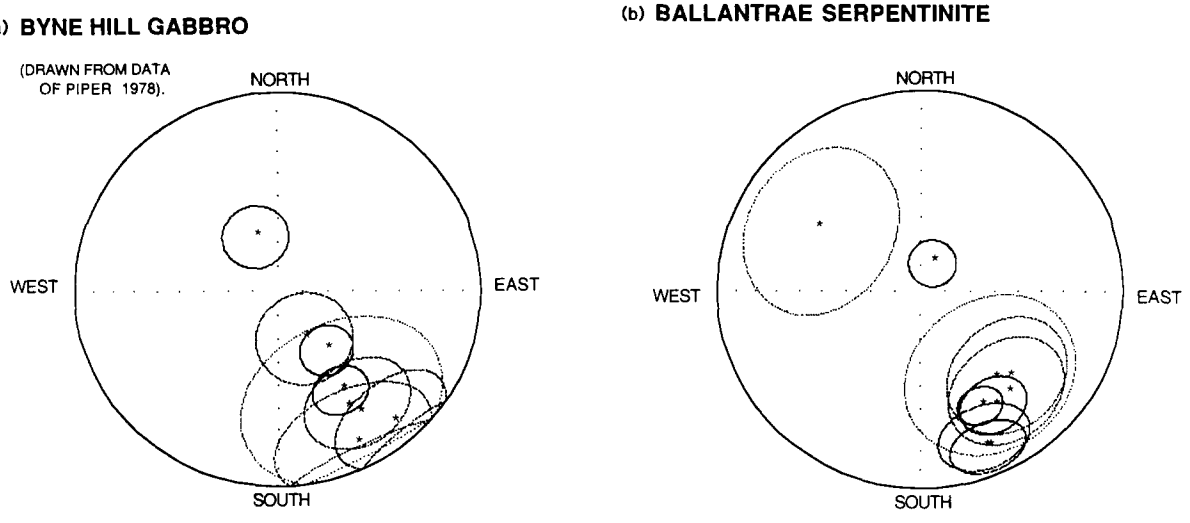


Fig. 10. Equal area projections show the site mean distributions of Piper [14] for the Byne Hill Gabbro (a) and Ballantrae northern belt serpentinite (b).

cant resolution of Tapetus during the Ordovician period. This problem arises due to the partial overlap of polar confidence circles determined in studies of Ordovician rocks either side of the Solway Line (see [35, fig. 1]).

6. Tectonic implications

The realisation that the characteristic magnetisation component (M) predates obduction related folding allows the calculation of a palaeolatitude. Furthermore, if remanence dates from the circulation of an early hydrothermal system; then the resulting latitude would approximate that of ophiolite formation.

Certain difficulties remain, however, in the statistical derivation of the latitude. Inappropriate calculation of a Fisher mean from a small circle partial arc centred on a vertical axis will result in a mean inclination that is artificially steep and consequently in a latitude that is incorrectly high [33]. This relationship is apparent in Table 2 where calculated Fisher means of strike and plunging fold corrected arcuate distributions yield palaeolatitudes of 34.5°S and 32.8°S, respectively. A more reliable estimate is given through one-dimensional analysis of inclination data only which yields a mean palaeolatitude of 28.8°S.

If the declination scatter following structural correction is attributed solely to a combination of true and apparent tectonic rotation about vertical axes, then the site mean results can be artificially constrained in declination (cf. [36]). An example of this is given in Table 2 producing a palaeolatitude confidence of $\pm 4.1°$. This result should be treated with caution, however, as the analysed distribution is necessarily non-Fisherian.

Unfortunately the absence of a well preserved sheeted dyke complex within the ophiolite precludes calculation of a palaeospreading orientation (cf. [37]). Nevertheless, the calculated palaeolatitude warrants comparison with existing Ordovician data from the Iapetus margins. A good agreement is evident when the result is compared to recently published palaeomagnetic data bordering the southern Laurentian margin ([38,39], both 22°S). Conversely, results are distinguishable from time equivalents derived from the Armorican massif (i.e. Hercynian Europe excluding southern Britain, cf. [40]) and Gondwana which are both estimated to lie in high southerly latitudes during Ordovician times [40].

With reference to the British database, the Ballantrae palaeolatitude lies within "Iapetus" as imaged by the 10° polar discordance in results to the north and south of the Solway Line [16,35]. Any resultant pole will not lie on the Ordovician segment of the British APWP, however, due to the sizable tectonic rotation encountered (approximately 80° clockwise when referred to contemporaneous British poles). This rotation may have been acquired either at the time of generation of the sequence, during ocean crust migration on to the Laurentian margin, or by subsequent rotation after terrane amalgamation.

The results from the northern belt serpentinites [14] are broadly concordant in declination with other British data and yield a palaeolatitude of 17.5°S. If this magnetisation relates to post-obduction serpentinisation then both a cessation of tectonic rotation and a palaeomagnetic record of northward obduction are implied.

Formation of the Ballantrae Ophiolite at a spreading axis close to the Laurentian margin is

TABLE 2

Summary of corrected magnetic directions and statistical analyses for the Slockenray Formation

Correction	N	D (°)	dD	I (°)	α_{95} (°)	P (°S)	dp	K
About strike	12	248.3	24.9	53.9	14.7	34.5	14.4	9.7
For plunging structure, $D = 290°$, $I = 80°$	12	231.2	22.0	52.2	13.5	32.8	12.7	11.3
Constrained declination	12	248.3	(7.1)	47.7	4.8	28.8	4.1	83.6
Inclination only, site	12	–	–	47.7	–	28.8	($\sigma = 8.9$)	–
Inclination only, sample	80	–	–	47.5	–	28.6	($\sigma = 11.5$)	–

N = number of sample sites; D, I = declination and inclination of site mean remanence vector; dD = uncertainty on the declination; P = palaeolatitude; dp = uncertainty on the palaeolatitude; K = Fisher precision parameter; σ = standard deviation for one-dimensional analysis.

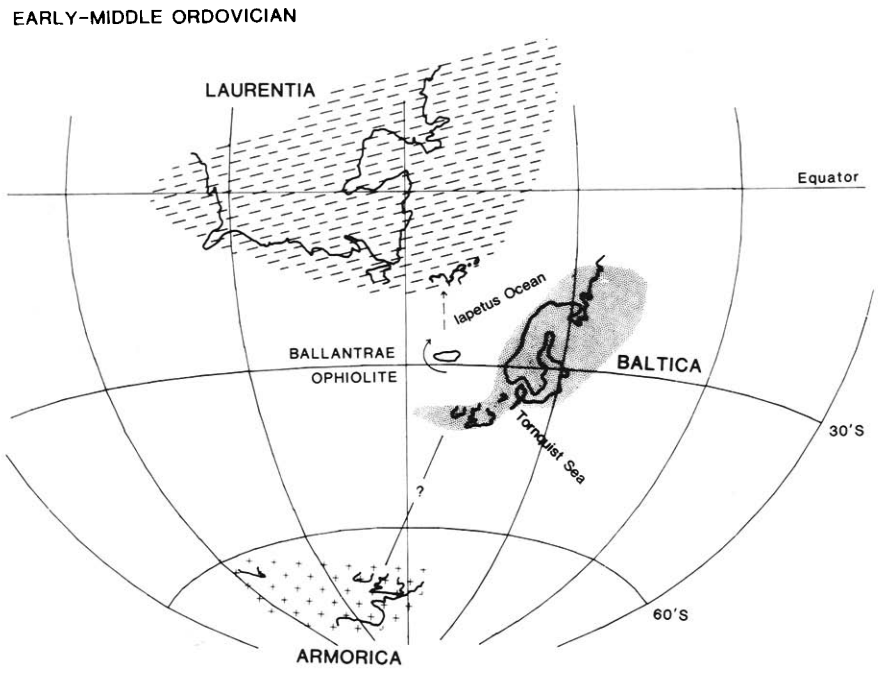


Fig. 11. An Early/Middle Ordovician palaeo-reconstruction for the Ballantrae Ophiolite and Iapetus bordering continents. Ophiolite rotation may take place either outboard, during migration or subsequent to terrane docking. The former positions of Laurentia, Baltica and Armorica are based on data from Briden, et al. [16,35] and Perroud et al. [40]. (Longitude is optional.)

required by the palaeolatitudinal data. The nature of such an axis has been the cause of differing opinion [4–6] and cannot be addressed by the present data. The latitudinal similarity between southern Laurentia and Ballantrae is consistent with the short-lived pre-obduction history of the ophiolite suggested by both radiometric and stratigraphic constraints however [10,41].

Finally, consideration of palaeolatitudinal data both north/south of the Solway Line and from Armorica suggests measurable separation between each block at least of a part of Ordovician times. The Southern British terrane (or "Eastern Avolonia", [42]) is separated from Laurentia by the Iapetus Ocean to the north and from Armorica by a further ocean to the south. The probability that southern Britain represents a far travelled microcontinental fragment which may have undergone rapid drift therefore becomes apparent. A fuller understanding of its role will await the reconciling of disparities between published apparent polar wander schemes for the period however (cf. [16,35,43,44]). This uncertainty in the former position of southern Britain is reflected in the palaeo-reconstruction shown in Fig. 11.

7. Summary and conclusions

The magnetic signature of both sediments and volcanics of the Slockenray Formation is multivectorial. Two components (S and M) have been identified differentiated on the basis of differing stability criteria. The less stable component S aligns (in situ) close to the present-day field direction and is considered to be of recent viscous origin. Component S occasionally displays higher stability both to thermal and AF cleaning, which is attributed to the presence of "hard" viscous remanence.

Component M forms the characteristic magnetisation of the formation and is resident in both magnetite and haematite. This component fails a conglomerate test on an intra-formational lava conglomerate and is identified as a remanence overprint. Remagnetisation may have taken place during circulation of an early hydrothermal system which effected magnetite resetting.

A magnetisation component of similar stability criterion to component M is found to display a near random distribution in the Benan Conglomerate of mixed provenance stratigraphically overlying the Slockenray Formation. This forms a positive conglomerate test and constrains a remagnetisation window for the characteristic magnetisation to a maximum of 30 Ma from Arenig to Llandeilo times.

The relation of component M to both folding and faulting suggests it to be pre-obduction in age. This allows the calculation of a palaeolatitude (28.8°S) for the complex during the Arenig which is consistent with a tectonic setting close to the southern Laurentian margin at the time of ocean crust generation.

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